

## Trainings ▾

### Preparatory Steps

#### One Piece Design, All Applications

- Remove the fuel pump, if necessary. Refer to Procedure 005-016 in Section 5.  
(/qs3/pubsys2/xml/en/procedures/18/18-005-016-tr.html)
- Remove the air compressor, if necessary. Refer to Procedure 012-014 in Section 12.  
(/qs3/pubsys2/xml/en/procedures/18/18-012-014-tr.html)



#### Two Piece Design, All Applications

##### **⚠ WARNING ⚠**

**Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.**



##### **⚠ WARNING ⚠**

**Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.**

- Disconnect the batteries. Refer to the OEM service manual.
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.

(/qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html)

- Remove the turbocharger oil drain line. Refer to Procedure 010-045 in Section 10.

(/qs3/pubsys2/xml/en/procedures/18/18-010-045-tr.html)

- Remove the lubricating oil filters. Refer to Procedure 007-013 in Section 7.

(/qs3/pubsys2/xml/en/procedures/18/18-007-013-tr.html)

- Disconnect the crankcase breather hose. Refer to Procedure 003-001 in Section 3.

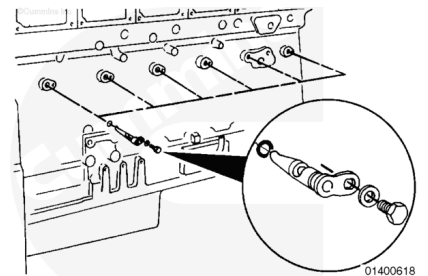
(/qs3/pubsys2/xml/en/procedures/18/18-003-001.html)

## Remove

### One Piece Design, All Applications

Remove the piston cooling nozzle capscrew.

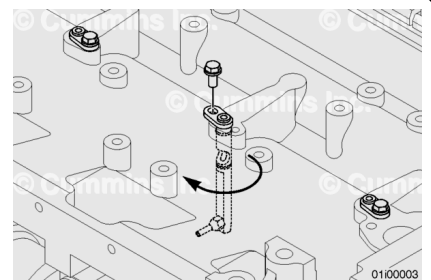
Remove the piston cooling nozzle.



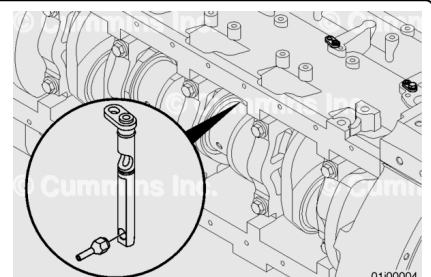
### Two Piece Design, All Applications

Remove the piston cooling nozzle capscrew.

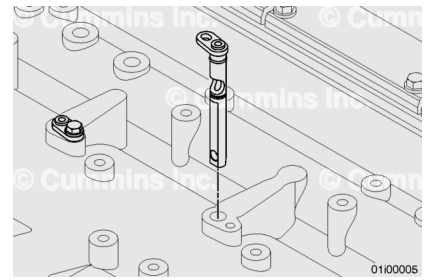
Rotate the nozzle body so the tip points toward the oil pan.



Remove the piston cooling nozzle tip.



Remove the piston cooling nozzle body.



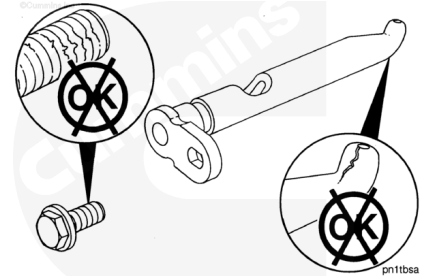
## Clean and Inspect for Reuse

### One Piece Design, All Applications

Clean the piston cooling nozzles.

Check for damage at the o-ring groove and spray tip.

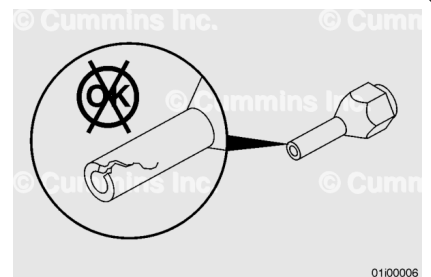
If the nozzle is damaged, it **must** be replaced.



### Two Piece Design, All Applications

#### **⚠ WARNING ⚠**

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



#### **⚠ WARNING ⚠**

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

#### **⚠ CAUTION ⚠**

Any damage to the piston cooling nozzle can result in major engine damage.

Clean the piston cooling nozzle tips with solvent and dry with compressed air.

Check the nozzle tips for damage.

If the nozzle is damaged, it **must** be replaced.

## Install

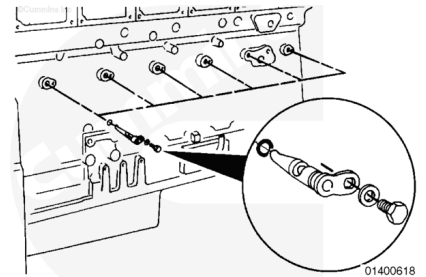
### One Piece Design, All Applications

Lubricate the o-ring with vegetable oil and install it onto the piston cooling nozzle.

Install the piston cooling nozzle, washer, and capscrew.

Tighten the capscrew.

**Torque Value:** 13 n•m [115 in-lb]



### Two Piece Design, All Applications

#### **⚠ CAUTION ⚠**

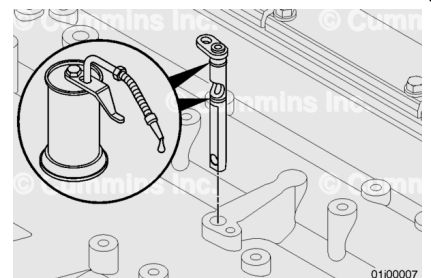
**Use care when handling the piston cooling nozzle. Any damage to the piston cooling nozzle can result in major engine damage.**

Lubricate the o-rings with vegetable oil.

Install the o-rings onto the piston cooling nozzle body.

Install the the nozzle body into the cylinder block with the tip opening pointing toward the oil pan.

The nozzle body should rest on the inboard (bottom) o-ring.

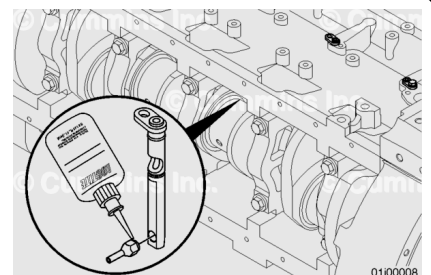


Apply Loctite™ 242, Part Number 3824041, onto the threads of the nozzle tip. Do **not** allow any Loctite™ to get inside the nozzle tip.

Install the nozzle tip onto the nozzle body.

Tighten the nozzle tip.

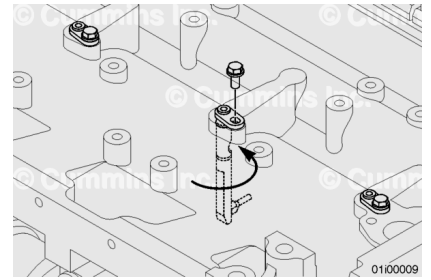
**Torque Value:** 9 n•m [80 in-lb]



Install the piston cooling nozzle the rest of the way into the cylinder block and rotate into position.

Install and tighten the mounting capscrew.

**Torque Value:** 14 n•m [124 in-lb]



## Finishing Steps

### One Piece Design, All Applications

- Install the air compressor, if removed. Refer to Procedure 012-014 in Section 12.  
(/qs3/pubsys2/xml/en/procedures/18/18-012-014-tr.html)
- Install the fuel pump, if removed. Refer to Procedure 005-016 in Section 5.  
(/qs3/pubsys2/xml/en/procedures/18/18-005-016-tr.html)



### Two Piece Design, All Applications

#### **⚠ WARNING ⚠**

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- Connect the crankcase breather hose. Refer to Procedure 003-001 in Section 3.  
(/qs3/pubsys2/xml/en/procedures/18/18-003-001.html)
- Install the lubricating oil filters. Refer to Procedure 007-013 in Section 7.  
(/qs3/pubsys2/xml/en/procedures/18/18-007-013-tr.html)
- Install the turbocharger oil drain line. Refer to Procedure 010-045 in Section 10.  
(/qs3/pubsys2/xml/en/procedures/18/18-010-045-tr.html)

- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html>)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html>)
- Connect the batteries. Refer to the OEM service manual.
- Operate the engine and check for leaks.

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**Last Modified: 30-Apr-2012**

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